$$
\begin{aligned}
&= 4\left(1+\frac{1}{3}\times\frac{1}{64}+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\times\left(\frac{1}{64}\right)^2+\cdots\right)\\
&= 4+\frac{1}{48}-4\times\frac{1}{9}\times\frac{1}{64}\times\frac{1}{64}+\cdots\\
&= 4+\frac{1}{48}-\frac{1}{9216}+\cdots\\
&\approx 4+0.02 \qquad \left(\text{since } \frac{1}{9216}+\cdots \text{ is very small}\right)\\
\sqrt[3]{65} &= 4.02 \text{ (approximately).}
\end{aligned}
$$

Example 5.25 Prove that $\sqrt[3]{x^3+7}-\sqrt[3]{x^3+4}$ is approximately equal to $\frac{1}{x^2}$ when x is large.

Solution:

$$
\begin{aligned}
\sqrt[3]{x^3+7} &= (x^3+7)^{\frac{1}{3}}\\
&= \left[x^3\left(1+\frac{7}{x^3}\right)\right]^{\frac{1}{3}}, \qquad \left(\left|\frac{7}{x^3}\right|<1 \text{ as } x \text{ is large}\right)\\
&= x\left(1+\frac{7}{x^3}\right)^{\frac{1}{3}}\\
&= x\left(1+\frac{1}{3}\times\frac{7}{x^3}+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{7}{x^3}\right)^2+\cdots\right)\\
&= x\left(1+\frac{7}{3}\times\frac{1}{x^3}-\frac{49}{9}\times\frac{1}{x^6}+\cdots\right)\\
&= x+\frac{7}{3}\times\frac{1}{x^2}-\frac{49}{9}\times\frac{1}{x^5}+\cdots\\
\sqrt[3]{x^3+4} &= (x^3+4)^{\frac{1}{3}}\\
&= \left[x^3\left(1+\frac{4}{x^3}\right)\right]^{\frac{1}{3}}\\
&= x\left(1+\frac{4}{x^3}\right)^{\frac{1}{3}} \qquad \left(\left|\frac{4}{x^3}\right|<1\right)\\
&= x\left(1+\frac{1}{3}\times\frac{4}{x^3}+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{4}{x^3}\right)^2+\cdots\right)\\
&= x+\frac{4}{3}\times\frac{1}{x^2}-\frac{16}{9}\times\frac{1}{x^5}+\cdots
\end{aligned}
$$

Since x is large, $\frac{1}{x}$ is very small and hence higher powers of $\frac{1}{x}$ are negligible. Thus $\sqrt[3]{x^3+7}\simeq x+\frac{7}{3}\times\frac{1}{x^2}$ and $\sqrt[3]{x^3+4}\simeq x+\frac{4}{3}\times\frac{1}{x^2}$. Therefore

$$
\begin{aligned}
\sqrt[3]{x^3+7}-\sqrt[3]{x^3+4} &\simeq \left(x+\frac{7}{3}\times\frac{1}{x^2}\right)-\left(x+\frac{4}{3}\times\frac{1}{x^2}\right)\\
\sqrt[3]{x^3+7}-\sqrt[3]{x^3+4} &= \frac{1}{x^2}, \text{ approximately}
\end{aligned}
$$